

PROBLEM SET / 03

Offline discovery from technical-document metadata

Offline technical-manual discovery across a controlled metadata boundary

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Developed after the original submission and grading dates. Status: In progress.

Abstract

Technical discovery has two distinct tasks: interpreting a scanned manual and searching what is permitted to leave the source workspace. Vintage Overseas Power Tools (VOPT) implements both as an offline pipeline. Original PDFs, OCR and evidence remain private to extraction and review; discovery receives a validated metadata snapshot with either coverage or approved values. [1]

The development corpus contains five admitted historical manuals and 616 pages. Layout-aware rules matched 13 of 17 selected reference fields, compared with four for a line baseline. Retrieval over reviewed values recorded recall@10 of 0.912 on 17 authored positive queries. These are development diagnostics with agent-inspected references, not held-out accuracy or human-gold estimates. [2] [3]

Problem and design objective

The SCADS task asks for technical attribute extraction from scanned documentation and natural-language document discovery. The implemented objective is discovery with inspectable constraints: identify the relevant document and model without inventing values or leaking source content into the released index. Numerical filtering requires released values; coverage-only metadata can establish that a topic is covered but cannot satisfy a threshold. [1]

Implemented contribution

A private extraction workspace, append-only review, strict release profiles and a separate discovery process form the core. The query compiler preserves model, variant, units and conditions. A guided Equipment -> Specification -> Results interface helps users express the bounded grammar while exposing unknowns and partial processing. [1] [4]

Evidence class: authentic historical scans used during development; agent-reviewed observations; authored queries; local packaging and interface verification. Source interpretation and retrieval are evaluated separately.

1. Research framing and information scope

DocLayNet addresses document-layout diversity with human-annotated pages across varied sources. Its motivation is relevant because success on a narrow family of manuals does not establish robustness to other layouts. Docling combines specialized layout and table models in a document-conversion toolkit. These are research precedents and possible comparators; the current VOPT implementation uses PyMuPDF, RapidOCR and explicit rules. [8] [9] [1]

Formal extraction and release objects

Represent an extracted assertion as $a=(d,r,m,v,t,x,u,k,e)$: document, revision, model, variant, attribute type, value/range, unit, condition and evidence. A candidate is not a full technical fact unless its scope and condition are supported. Let $R_{\pi}(A)$ be the approved release under policy π ; discovery receives only $C=R_{\pi}(A)$, not A or the source PDFs. [1] [10]

coverage profile: document/model + reviewed attribute presence
values profile: coverage + approved values/units/conditions
search(q,C): validate q ; apply exact predicates; rank scopes
explanation(result): use released catalog fields only

The key boundary can be stated as an invariance property: for two private workspaces producing the same released catalog C , a deterministic discovery query q must return the same result. This is a proposed formal interpretation of the implemented isolation tests. It constrains hidden use of OCR or private values, while not claiming a proof against every side channel. [5] [7]

Research questions

RQ1 asks whether complete model-attribute-value-condition assertions can be extracted reliably from representative scans. RQ2 asks whether approved metadata satisfies discovery needs with hard-constraint correctness. RQ3 asks whether release changes remove information from all search paths. These require separate denominators: excellent retrieval cannot repair incorrect extracted facts, and deliberate policy omission is not an extraction error.

2. Method and implemented architecture

Workspace	Implemented method	Output
Extraction	PyMuPDF geometry/native text; local RapidOCR ONNX for image pages	Page records, model-scoped candidates and evidence coordinates
Review	Append-only decisions bound to extraction fingerprints	Accepted/corrected assertions with reviewer provenance
Release	Field allowlists and separate coverage/value profiles	Validated versioned metadata bundle
Discovery	SQLite, exact predicates and BM25; TF-IDF comparator	Ranked document/model scopes and request references

Every candidate must preserve document revision, model, attribute, value or range, unit and rating/operating qualifiers. A correctly read number can still be attached to the wrong component or configuration. Review therefore examines factual scope rather than treating OCR confidence as a probability that a specification is true. [1] [2]

Coverage is a separate product

Coverage exports describe verified attribute presence. Value exports may additionally include approved typed values and conditions. A coverage query such as which manuals discuss voltage differs from a request for equipment above a voltage threshold. The discovery service rejects unsupported numeric filtering under a coverage-only profile rather than dropping the constraint. [1] [5]

Exact predicates precede ranking

The query plan is inspectable: identifiers, attribute roles, operators, units, Boolean structure and conditions are explicit. Hard predicates enforce same-model and compatible-variant scope before ranking. BM25 is the default ranker. The semantic comparator is ontology-normalized TF-IDF, not a pretrained embedding system. Both share the same predicate compiler. [1] [3]

Atomic catalog import and full index replacement support controlled release changes. Review and extraction fingerprints detect stale decisions, while monotonic withdrawal handling prevents a rollback from simply restoring removed metadata. The separate discovery installation has neither source files nor extraction dependencies in its verified profile. [1] [5] [7]

3. Scoped extraction and query evaluation

The pipeline uses page geometry and local OCR, then applies model/header and unit rules. Per-page state distinguishes processed, failed and unknown material. Evidence review is bound to source and extraction fingerprints so that a prior decision cannot silently approve changed source bytes. [\[1\]](#) [\[7\]](#) [\[10\]](#)

```
for each admitted document revision:
  verify source identity; process bounded pages/checkpoints
  recover text/layout; retain page and region coordinates
  emit candidates only with supported model/attribute scope
  normalize units without changing physical attribute role
  record original candidate plus explicit review events
  export only approved policy fields as a versioned bundle
  import atomically into the source-free discovery database
```

Conjunction is not a document-level keyword join

For a model scope s , a conjunctive request requires all predicates to have compatible support in that scope. A document containing model A at 230 V and model B at 2 kW cannot automatically satisfy a request for one model having both properties. Operating conditions must also align. Discrete alternatives are not continuous ranges, and an unresolved qualifier prevents an unjustified positive match. [\[1\]](#) [\[3\]](#) [\[10\]](#)

```
match(s, q1 AND q2) requires compatible support for both
unknown attribute -> unknown; not zero and not false
coverage-only + numeric constraint -> unsupported
ambiguous physical role -> clarify before ranking
```

Ranking is downstream of these constraints. A lexical or semantic similarity score may order valid scopes but cannot override an unsupported numeric predicate. This preserves meaning even when user language is abbreviated. The guided UI compiles the same query contract rather than maintaining a separate permissive search path. [\[3\]](#) [\[4\]](#)

Pipeline cost separates page rendering/OCR, extraction, review and metadata indexing. Synthetic catalog timings measure the last stage only. A deployment claim should specify page count, scan resolution, layout, OCR engine, retry policy, memory and review effort rather than scaling catalog-query speed into document throughput.

4. Worked example

A verified browser journey asks for model HR-28 with output power at least 1.4 kW, output voltage at least 30 V and DC operation. The guided interface returns one confirmed match in the Homelite manual, along with the partial-processing notice. Supporting assertion IDs and the source revision remain attached to the request reference. [4]

User intent	Interpretation	Observed behavior
Generator manuals covering output voltage	Coverage; no threshold	M7 and HR-28 confirmed; incomplete scopes shown separately.
Output voltage at least 125 V	Value constraint	No confirmed matches; six records require more detail.
HR-28, >=30 V, >=1.4 kW, DC	Same model and compatible conditions	One confirmed match with provenance and partial-processing notice.
Voltage above 100 V	Input/output role ambiguous	Clarification required.

This example illustrates why model and qualifier association matter more than matching a keyword. Input voltage is not output voltage, and electrical input power must not become mechanical output power through a unit conversion. A compendium containing several models cannot satisfy a conjunction by joining one model's voltage to another model's power. [1] [4]

A request, not an invented document answer

The exported JSON stores a document/version reference, catalog identity, query and supporting assertion IDs. A verified journey saved HR-28 from a coverage query, then ran a different numeric query; the actual downloaded request still retained the originating query. It contained released metadata rather than original PDF or OCR content. [4]

Unknowns remain visible

An unobserved field means unknown, not absent or zero. Partial page processing is retained with matches, and ambiguous operating conditions are not converted into broad ranges. A discovery result therefore describes what the approved catalog can support, while document acquisition remains a separate action. [1] [3] [4]

Table 1. These are agent-operated browser checks on the frozen development catalog. They verify specific interface behavior, not independent usability or generalized extraction accuracy. [4]
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5. Recorded results and interpretation

Measure	Observed value	Denominator / scope
Admitted scans	5 manuals; 616 pages	610 processed pages; 6 explicit failures. [2]
Selected field matches	Layout 13/17; line 4/17	Document, page, model, attribute, unit, value and qualifier. [2]
Reviewed-values retrieval	Macro recall@10 0.912; macro MRR@10 0.941	17 positive authored queries; 22 known scopes. [3]
Reference-values retrieval	Macro recall@10 1.000; macro MRR@10 1.000	Same known positives; diagnostic reference metadata. [3]
Plan status checks	33/33	Authored source-reference queries. [3]
Recorded regression suite	151 passed	Build-verification snapshot; separate from later UI checks. [5]

Selected field matches are not exhaustive recall or full-assertion precision. The extraction report notes that conditions and component meaning need additional review. The retrieval reference condition is also deliberately separate from reviewed extraction: perfect reference-metadata retrieval cannot conceal omissions in the actual catalog. Both rankers produced equal reported results in these development comparisons. [2] [3]

An informative failure distinction

A preserved first M18 audit supported all 17 numeric transcriptions but only five candidates with essential factual scope. Nine needed scope repair and three named the wrong entity. A repaired seven-candidate record was source-inspected as supported, but suppressed observations and a known development manual prevent a recall or generalization claim. The lesson is that number recognition and technical truth are different evaluation targets. [2]

Offline evidence

The recorded portable discovery and extraction profiles cold-started on the existing Windows host using packaged assets, with Python network attempts blocked. Discovery operated without private source files. These checks support the tested packaging boundary; they are not OS-level network isolation or independent clean-target certification. [5]

6. Proposed independent evaluation

PROPOSED STUDY. Independent labels and untouched families are a prospective validation design. Existing development matches and agent reviews must not be reused as final-test evidence.

Track	Unit and reference	Measure
Extraction	Complete source-supported assertion; exhaustive human annotation	Precision/recall including model, variant, unit and condition
Evidence location	Page/region supporting the full assertion	Localization correctness and review time
Reference catalog search	Independent queries and adjudicated relevant scopes	Recall@k, reciprocal rank and hard-constraint errors
Extracted catalog search	Same queries with reviewed and automatic conditions separated	Downstream loss attributable to extraction
Release lifecycle	Fixed-source catalogs under downgrade/withdrawal/rollback	Invariance, removed-field and stale-index checks

Partition manuals by equipment family and related edition before tuning. Independent annotators should capture all supported assertions in sampled pages, including negative and ambiguous cases, rather than selecting only promising leads. Adjudicate component identity, rating and conditions. Freeze queries and threshold choices before opening final families. [7] [11]

Use exact-match scoring on the full assertion tuple, with separately reported numeric transcription and unit normalization diagnostics. For retrieval, labels must distinguish information in the original manual from information actually released by a profile. Compare lexical and semantic rankers on the same candidate scopes and hard constraints; use paired query-level uncertainty estimates and report unjudged results.

Operational validation should cold-start on an actual target with networking unavailable, original sources inaccessible to discovery and packaged model assets verified. Inject interruption, changed-source, malformed import, downgrade and revocation events. Measure review effort against manual cataloging at comparable factual quality before claiming labor reduction.

7. Roadmap and scientific interpretation

Phase	Output	Decision rule
Qualify diversity	Rights-qualified corpus stratified by layout, scan and model scope	Freeze representative families and excluded rights cases.
Create independent references	Exhaustive assertions, relevance labels and adjudication	Keep human and agent provenance separate.
Run controlled studies	Raw/reviewed/reference extraction-search comparisons	Choose methods by full-scope quality and effort.
Qualify deployment	Target-machine offline lifecycle and package inventory	Reproduce exact scope and failure behavior.

The implementation already supports private extraction, review, controlled release and metadata search. The proposed roadmap expands its research qualification. A future local model should be introduced only when measured rule failures justify its cost, with the same evidence and release constraints. The current TF-IDF comparator provides no basis for claiming pretrained semantic retrieval benefits. [1] [3]

Threats to validity

The historical equipment corpus is narrow, and every family was inspected during development. Agent-selected leads favor observable fields and do not exhaustively enumerate source facts. Damaged pages create unknown coverage. Corrected review records improve a catalog but change the estimand: retrieval over reviewed data is not raw extraction accuracy. [2] [3]

Policy itself affects relevance. A manual can contain a value that discovery correctly refuses to expose. Scoring that refusal as retrieval failure would conflate intended information control with an engineering defect. Conversely, a high search score cannot authorize release of a private field. Evaluation should therefore report both source relevance and profile-visible relevance.

Conclusion

VOPT demonstrates an offline workflow in which source interpretation and disseminated discovery are distinct, auditable stages. Development evidence shows why full technical scope, not numeric transcription alone, is the appropriate research unit. The next scientific step is independent, family-separated evaluation of that unit and its effect on document discovery.

8. Reproducibility and research record

Use the operating guide for the private extraction/review and released discovery workflows. Frozen evaluation records preserve the processing versions that produced them; do not silently reprocess them and present new results as the original experiment. Source and review provenance are part of the research artifact. [6] [7]

```
python -m vopt doctor
python -m pytest -q
python scripts/evaluate_extraction.py --skip-ingest
```

Run these commands from the environment described in the implementation guide and RUNBOOK. Exact historical extraction replay requires the separately retained frozen runs/real-workspace or research-evidence bundle; a Git-only checkout is not a self-contained copy of that workspace. Fresh corpus acquisition and OCR constitute a new run. For discovery reproduction, use a versioned released catalog and inspect its query plan. Package hashes and later UI checks identify their respective snapshots. [1] [4] [5] [6]

Publication reproduces recorded measurements, not a new OCR evaluation. No human-gold, untouched-family, labor-savings or classified-deployment claim is made.

A reproducible research record

The public nested project source contains implementation, documented contracts and retained development evidence. The publication evidence manifest records SHA-256 fingerprints for each local source cited in this paper. External references link to primary publications or official technical documentation, verified for this edition.

A future rerun should record the exact commit, dependency/runtime identities, input hashes, configuration and environment, plus each attempt and its outcome. Keep generated outputs separate from source and mark any changed implementation as a new condition. Reproduction of a software check does not automatically reproduce a model judgment, benchmark distribution or independent human label.

Authorship and project status

Boomer Rawlings developed this work individually after the original submission and grading dates. The current project is In progress. The implemented methods and recorded observations in this paper are separated from the explicitly proposed research protocol and roadmap. No group-project, graded-submission or completed-research claim is made.

References

Repository references identify the evidence used for the implementation and result claims. Primary literature supports the research framing; its published results are not attributed to this project.

[1] [Implemented system and architecture](#)
IMPLEMENTATION.md

[2] [Extraction development diagnostics](#)
reports/extraction-development.md

[3] [Retrieval development evaluation](#)
reports/retrieval-development.md

[4] [Guided interface verification](#)
reports/interface-verification.md

[5] [Build and portable-runtime verification](#)
reports/build-verification.json

[6] [Operating and reproduction guide](#)
RUNBOOK.md

[7] [Measured acceptance and evidence scope](#)
reports/acceptance-status.md

[8] [Pfitzmann et al. \(2022\). DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. KDD; arXiv:2206.01062.](#)
<https://arxiv.org/abs/2206.01062>

[9] [Auer et al. \(2024\). Docling Technical Report. arXiv:2408.09869.](#)
<https://arxiv.org/abs/2408.09869>

[10] [Project query, release and evidence contracts](#)
docs/implementation-contract.md

[11] [Frozen evaluation protocol](#)
eval/protocol.md